Exercise 18: Gated Clocks

- You will modify a multiply-accumulate unit to use gated clocks

You can work in pairs for this exercise

- Download the following 2 files:
  - mult_accum.sv → design that uses recirculating flops
  - mult_accum.dc → synthesis script
- Synthesize **mult_accum.sv** using the provided script
  - Note the area, and the power estimate
  - Note whether or not it passes timing

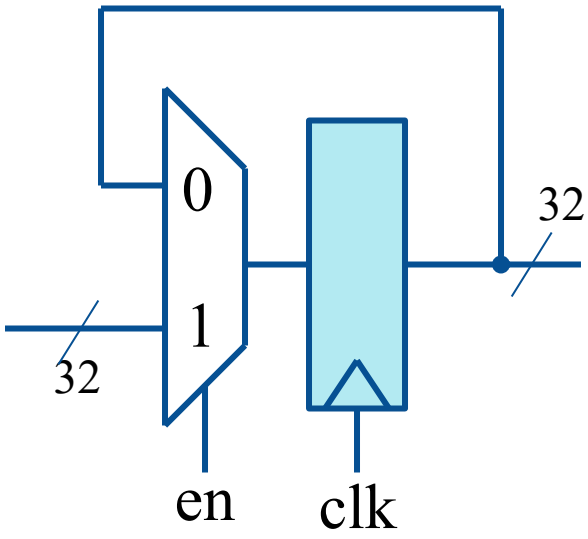
- Now modify the design to use gated clocks instead of recirculating.
- Modify the synthesis script as needed.
- Again synthesize and remark on the difference in both area and timing.

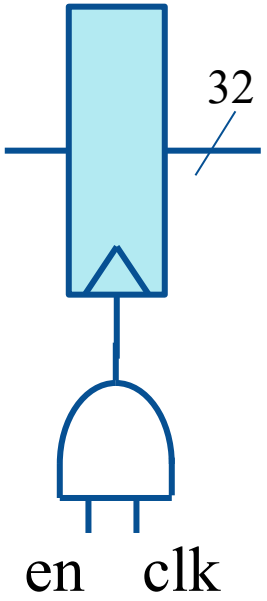
- **Submit**
  - Your modified design **mult_accum_gated.sv**
  - A table showing the area and dynamic power numbers for both designs.

Hints

- On a scrap paper, draw how the provided (non clock gated) version of the code would synthesize
  - Use clouds for combinational, a 2:1 mux for recirculating mux, and simple flops.
  - There are 3 **always_ff** blocks:
    - One is enabled and produces **prod_reg**. It is 16-bits wide. Definitely a candidate for clock gating.
    - One is free running and simply delays **en** to produce **en_stg2**. Definitely **NOT** a candidate for clock gating.
    - One is enabled and produces **accum.** It is 64-bits wide. Definitely a candidate for clock gating.
- So you should have 2 gated clocks (one for **prod_reg** and another for **accum**). Note that **accum** has a synchronous clear (**clr**).
  - For the clear to occur the clock has to be running when **clr** is high. **en_stg2 | clr**

Gated Clocks Review

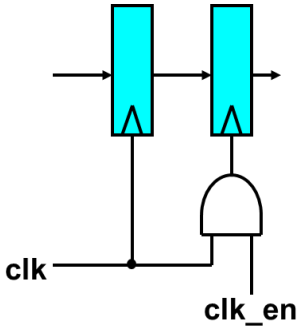




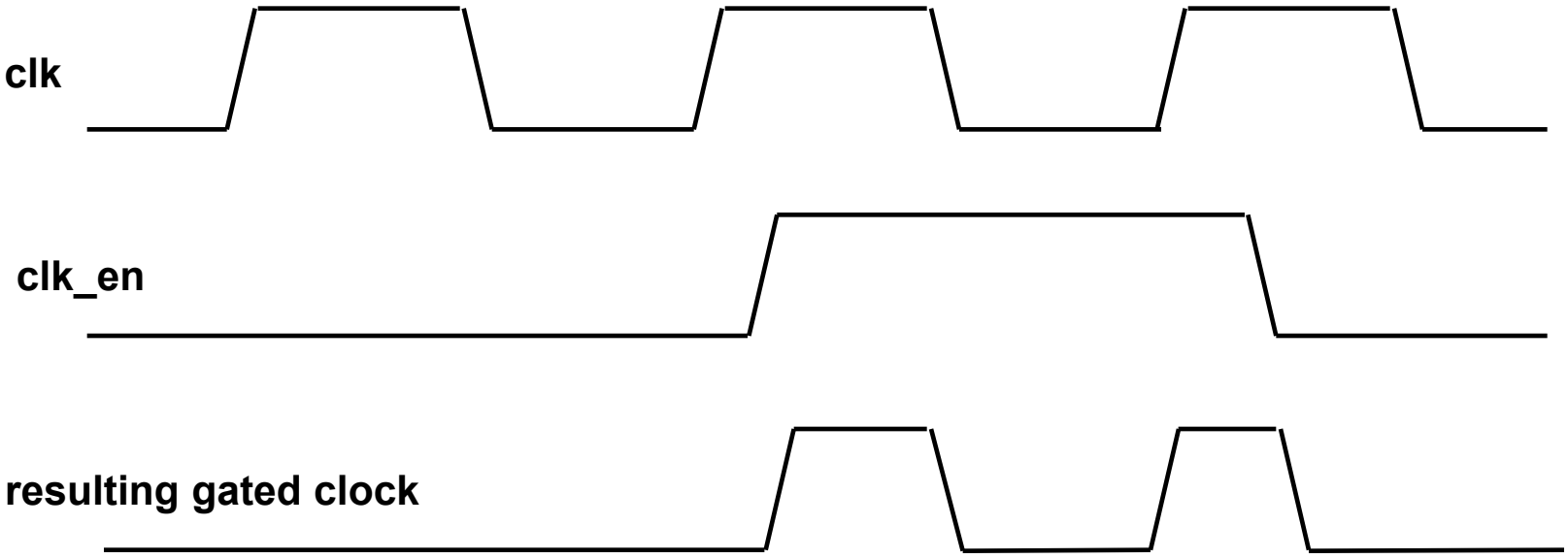
Recirculating seems rather wasteful we need a 2:1 mux for each flop. Plus all the cap and gates inside the flops on the clock network will be toggling.

Wouldn't it be better to just gate the clock? Pretty much same functionality but should save area and power? Yes, but you have to do it right.

Gated Clocks Review

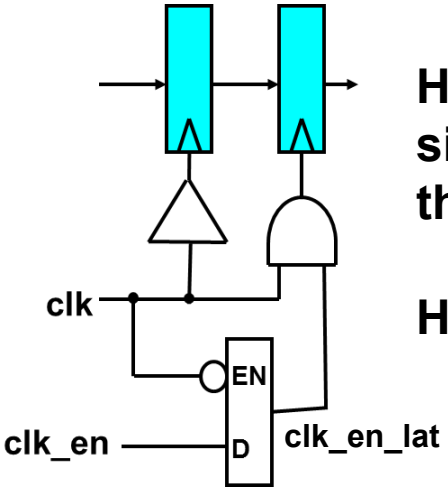


Something this simple does not work. Recall the enable function is probably coming from a SM. So it changes slightly after the rising edge of clock.



That's pretty messed up!

Gated Clocks Review



Have to use an enable low latch to latch the clock enable signal. This ensures the enable is setup and ready for the next high phase of clock

Have to balance from clock root to all leaf nodes of clock

